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**Course:** Calculus Beta

**Assignment:** MyLab Math - Uge 13

Find all the local maxima, local minima, and saddle points of the function.

$$f(x,y) = 4xy - x^2 - 9y^2 + 6x + 10$$

The local extrema and saddle points in the interior of the domain of a function occur at critical points of the function. An interior point of the domain is a critical point if both  $f_x$  and  $f_y$  are zero or if one or both derivatives do not exist.

Find the partial derivative with respect to  $x$ .

$$f_x = 4y - 2x + 6$$

Find the partial derivative with respect to  $y$ .

$$f_y = 4x - 18y$$

Both derivatives exist everywhere in the domain of  $f(x,y)$ . Set each derivative equal to zero and solve the system of equations for  $x$  and  $y$ .

$$4y - 2x + 6 = 0$$

$$4x - 18y = 0$$

Multiply the top equation by 2 and solve using elimination. Add the two equations to eliminate  $x$ .

$$\begin{array}{r} 8y - 4x + 12 = 0 \\ -18y + 4x = 0 \\ \hline -10y + 12 = 0 \end{array}$$

Solve for  $y$ .

$$\begin{array}{l} -10y + 12 = 0 \\ y = 1.2 \end{array}$$

Find the corresponding value of  $x$ . Substitute 1.2 into the second equation.

$$4y - 2x + 6 = 0$$

$$4(1.2) - 2x + 6 = 0$$

$$x = 5.4$$

There is only one critical point for  $f(x,y) = 4xy - x^2 - 9y^2 + 6x + 10$  and it occurs at  $(5.4, 1.2)$ .

The nature of the critical point can be tested using the second derivative test if the first- and second-order partial derivatives of  $f$  are continuous throughout a disk centered at point  $(a,b)$  and  $f_x(a,b) = f_y(a,b) = 0$ .

i.  $f$  has a local maximum at  $(a,b)$  if  $f_{xx} < 0$  and  $f_{xx}f_{yy} - f_{xy}^2 > 0$  at  $(a,b)$ .

ii.  $f$  has a local minimum at  $(a,b)$  if  $f_{xx} > 0$  and  $f_{xx}f_{yy} - f_{xy}^2 > 0$  at  $(a,b)$ .

iii.  $f$  has a saddle point at  $(a,b)$  if  $f_{xx}f_{yy} - f_{xy}^2 < 0$  at  $(a,b)$ .

iv. The test is inconclusive at  $(a,b)$  if  $f_{xx}f_{yy} - f_{xy}^2 = 0$  at  $(a,b)$ .

The expression  $f_{xx}f_{yy} - f_{xy}^2$  is called the discriminant. Find each derivative in the expression. Differentiate  $f_x$  with respect to  $x$  to find  $f_{xx}$ .

$$f_x = 4y - 2x + 6$$

$$f_{xx} = -2$$

Differentiate  $f_y$  with respect to  $y$  to find  $f_{yy}$ .

$$f_y = 4x - 18y$$

$$f_{yy} = -18$$

Differentiate  $f_x$  with respect to  $y$  to find  $f_{xy}$ .

$$f_x = 4y - 2x + 6$$

$$f_{xy} = 4$$

Evaluate the discriminant at  $(5.4, 1.2)$  to determine the nature of the critical point at that point.

$$\begin{aligned}f_{xx}f_{yy} - f_{xy}^2 &= (-2)(-18) - (4)^2 \\&= 20\end{aligned}$$

At the point (5.4,1.2), the discriminant is 20 and  $f_{xx}$  is  $-2$ . Therefore, (5.4,1.2) is a local maximum.

To find the maximum value at that point, evaluate the function at the local maximum.

$$\begin{aligned}f(5.4,1.2) &= 4(5.4)(1.2) - (5.4)^2 - 9(1.2)^2 + 6(5.4) + 10 \\&= 26.2\end{aligned}$$

Therefore, there is a local maximum at (5.4,1.2) with a value of 26.2.

Since there was only one critical point, there are no local minimums and no saddle points.